

Boundary Conditions of the Third Kind: Generalized Sources and Sinks in Fluid Simulations

Jeff Lait
Side Effects Software Inc.
Canada
jmlait@gmail.com

Omar Zarifi
Side Effects Software Inc.
Canada
omarz@sidefx.com

Tomáš Skřivan
Carnegie Mellon University
United States of America
skrivan.tomas@gmail.com

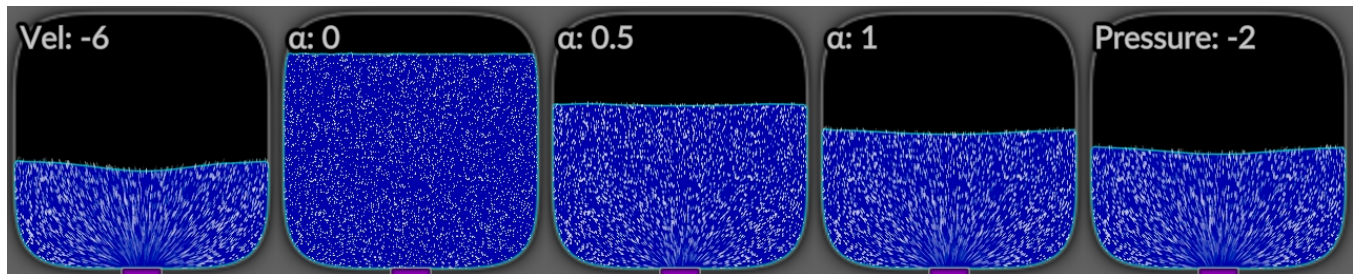


Figure 1: Fluid draining with different velocity, Robin, and pressure boundary conditions.

Abstract

Fluid simulations use Dirichlet or Neumann boundary conditions to create free surfaces, colliders, source, and sinks. There is, however, a third condition: the Robin boundary condition. Considered relevant only for heat or electromagnetic problems, we found it provides a consistent framework to replace the binary choice of Dirichlet or Neumann with a continuum. Blending these conditions enables the simulation of real-world fluid behaviours otherwise not achievable.

CCS Concepts

• **Mathematics of computing** → Solvers; • **Applied computing** → Physics.

Keywords

fluid simulation, boundary conditions, Robin boundaries

ACM Reference Format:

Jeff Lait, Omar Zarifi, and Tomáš Skřivan. 2024. Boundary Conditions of the Third Kind: Generalized Sources and Sinks in Fluid Simulations. In *SIGGRAPH Asia 2024 Technical Communications (SA Technical Communications '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3681758.3697988>

1 Introduction

Liquid simulations for computer graphics assume pressure waves are dissipated over a timestep and thus use a projection method to directly find an equilibrium state [Stam 1999]. The closest divergent

free velocity field is found by computing a pressure field that would remove divergence when applied as an impulse. This results in a Poisson problem whose solution is highly dependent on boundary conditions.

Air-liquid surfaces are given a Dirichlet boundary enforcing a zero pressure on the air side; while air-solid surfaces are given a Neumann boundary enforcing a zero pressure gradient across the boundary. The importance of boundary conditions cannot be overstated. By tweaking boundary conditions, one can improve the flow along the solid boundary with variational methods [Batty et al. 2007], or support surface tension or bubbles by varying the air pressure [Goldade et al. 2020].

There is a third boundary condition which enforces an affine relationship between pressure and pressure gradient: the Robin boundary.

2 Sources and Sinks

Air, solid, and liquid suffices for closed systems, but many dynamic simulations have continuous inflow and outflow. A river simulation must create water at the inlet and destroy it at the outlet. A patch of water in a wave tank must destroy splashes that leave, but also create water to flow back in and replenish the tank.

A simple sink or source is a volume that defines where fluid should be destroyed or created. New fluid is assigned an initial velocity from the source. This method lacks realism: a source will stop filling when submerged and back-pressure can reverse inlets.

Flux Animated Boundary (FAB) conditions [Stomakhin and Selle 2017] addressed these deficiencies. A source volume generates a collider-like boundary condition with the wall-velocity set to the fluid's flux, rather than the wall's motion. The enforced velocity makes it simple to compute fill rates and artistically adjust the source. Submerged sources will continue to emit at the same rate.

FAB can be generalized by fixing the pressure boundary, forming a Pressure Animated Boundary (PAB) [Side Effects Software Inc. 2023]. When a source is submerged its rate of emission drops as

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

SA Technical Communications '24, December 03–06, 2024, Tokyo, Japan

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1140-4/24/12

<https://doi.org/10.1145/3681758.3697988>

hydrostatic pressure increases, eventually reaching equilibrium. A tank with hydrostatic pressure gradient on the walls will maintain a constant water level.

Table 1: Source and Sink Taxonomy

Type	Fluid	Velocity	Pressure
Pure Source	Create		
FAB Source	Create	Enforce	
PAB Source	Create		Enforce
Pure Sink	Destroy		
FAB Sink	Destroy	Enforce	
PAB Sink	Destroy		Enforce
Collider		Enforce	
Air			Enforce

We can compose these types into four volumes: creation, destruction, velocity boundary, and pressure boundary. But can the two boundary types be unified?

3 Physical Meaning

Physical forms of the Robin boundary condition are usually found in heat-flow [Miller et al. 2024] and electromagnetic problems. Heat was once considered a fluid, and there are strong relationships between fluids and electrostatics [Lait 2018], so we should expect liquids to admit an interpretation.

Real-world sinks can drain at different speeds depending on the state of the plumbing. Given a pressure-driven sink with zero pressure, a negative pressure will speed up the sink, but a positive pressure will not suffice to slow it; the sink will instead stop draining when equilibrium is reached. The Robin boundary allows a single cell to have an implicit porous microstructure on its boundary.

Wave tanks are another use. PAB allows tanks to re-fill, but lacks boundary velocities. FAB allows velocity specification, but cannot recover from volume changes. A Robin-based blend of the two creates a leaky pump, allowing an artist to balance the virtues of both.

4 Discrete Robin Boundary Derived

Our Poisson problem's variable is pressure, p . Expressing the Robin boundary in these terms, with Δp taken on the boundary's normal:

$$\alpha p + \beta \Delta p = \gamma \quad (1)$$

Since $\alpha = 1$ and $\beta = 0$ recovers the Dirichlet condition, and $\alpha = 0$ and $\beta = 1$ recovers the Neumann condition, this condition subsumes the two boundary types.

4.1 Pressure Matrix

The standard pressure projection procedure is:

- (1) $u' = \text{enforcevel}(u)$
- (2) $d = \text{div}(u')$
- (3) $A = \text{div}(\text{grad}(\cdot))$
- (4) $p = A^{-1}d$
- (5) $u'' = u' - \text{grad}(p)$

Table 2: 1D MAC grid

p_-	p_0	p_+
-------	-------	-------

d is the fluid's original divergence. A is the operator converting a discretized pressure array into a divergence array.

We use a MAC grid discretization: velocities are on the faces; pressure and divergence are on the centers of cells. A uniform and isotropic discretization eliminates grid size from the equations.

A can be constructed as a symmetric positive definite (SPD) sparse matrix. A should map a vector with a pressure value per-cell to a vector with a divergence per-cell. For the MAC discretization, we can compute this directly. The 1D case will be worked through, higher dimensions generalize in a straightforward manner.

We focus on the cell labeled 0. The cell in the negative and positive directions are labeled $-$ and $+$. Velocities are stored on faces so there are two velocities: one measuring flux between $-$ and 0 labeled $-$ and one between 0 and $+$ labeled $+$.

The cell's divergence comes from the boundary velocities:

$$d_0 = -u_- + u_+ \quad (2)$$

The boundary's velocity comes from the pressure gradient:

$$u'_- = u_- + p_0 - p_- \quad (3)$$

$$u'_+ = u_+ + p_+ - p_0 \quad (4)$$

We can compute divergence directly from pressure:

$$d_0 = -(p_0 - p_-) + (p_+ - p_0) \quad (5)$$

$$d_0 = -2 * p_0 + p_+ + p_- \quad (6)$$

This gives the canonical form of A : a diagonal of -2 and 1 in off diagonals where fluid cells connect.

4.2 Pressure Boundaries

We will consider only the case where p_+ is a boundary condition. The same argument can apply, up to signs, for p_- .

To make the $0+$ boundary a free surface, we enforce the p_+ value to a fixed surface pressure, q . q is usually zero for air, but will vary for pressure animated boundaries.

$$p_+ = q \quad (7)$$

$$d_0 = -2 * p_0 + q + p_- \quad (8)$$

q is constant so can be absorbed into the goal divergence, d .

4.3 Velocity Boundaries

To set a specific velocity on the $0+$ boundary, we can pre-enforce by explicitly setting u_+ to the goal velocity. The requirement in the pressure solve is to not further change the boundary velocity. From (4), we require:

$$p_+ = p_0 \quad (9)$$

$$d_0 = -2 * p_0 + p_0 + p_- \quad (10)$$

$$d_0 = -p_0 + p_- \quad (11)$$

The matrix formulation acts as if the other cell does not exist; the connection is lost from the diagonal and the off-diagonal.

Pre-enforcing the velocity makes it hard to unify the conditions. Instead we can determine the impulse j_+ required to reach our goal velocity. To update u_+ by j_+ , we can use (4):

$$u'_+ = u_+ + p_+ - p_0 = u_+ + j_+ \quad (12)$$

$$p_+ = j_+ + p_0 \quad (13)$$

$$d_0 = -2 * p_0 + (j_+ + p_0) + p_- \quad (14)$$

$$d_0 = -p_0 + p_- + j_+ \quad (15)$$

j_+ is constant, like q for pressure, so can be absorbed into d .

4.4 Velocity Update

After solving the pressure vector, we update velocity with (3) and (4). For our boundary conditions, A did not solve for p_+ . To do the velocity update (4), p_+ must be computed using (7) or (13).

$$u'_+ = u_+ + q - p_0 \quad (16)$$

$$u'_+ = u_+ + j_+ \quad (17)$$

(16) is the velocity update for a pressure boundary, and (17) for a velocity boundary.

4.5 Robin Boundaries

We can now derive the Robin boundary. Focusing on the 0+ boundary, we write (1) in those terms:

$$\alpha p_+ + \beta(p_+ - p_0) = \gamma \quad (18)$$

$$\alpha p_+ + \beta p_+ = \gamma + \beta p_0 \quad (19)$$

$$p_+ = (\gamma + \beta p_0) / (\alpha + \beta) \quad (20)$$

There is limited freedom in the Robin constants. If $\alpha + \beta = 0$, we cannot solve for p_+ . We take the ansatz of $\alpha + \beta = 1$. Then, with a suitable γ' , we reduce to two variables.

$$p_+ = \gamma' + \beta p_0 \quad (21)$$

We note this is a pure generalization of (7) and (13) Computing d_0 to determine the A coefficients:

$$d_0 = -2 * p_0 + (\gamma' + \beta p_0) + p_- \quad (22)$$

$$d_0 = -(2 - \beta) * p_0 + p_- + \gamma' \quad (23)$$

γ' is constant and can absorb into d . The diagonal of A is adjusted by the value of β .

Any Robin boundary condition can now be solved with suitable γ' and β . We can determine Robin values for pressure and velocity conditions, so we re-introduce α as the convex mix of (7) and (13):

$$p_+ = \alpha q + (1 - \alpha)(p_0 + j_+) \quad (24)$$

$$p_+ = \text{lerp}(j_+, q, \alpha) + (1 - \alpha)p_0 \quad (25)$$

$$\gamma' = \text{lerp}(j_+, q, \alpha) \quad (26)$$

$$\beta = (1 - \alpha) \quad (27)$$

Because j_+ , q , and α span γ' and β , we can conclude that any Robin boundary condition can be expressed as a blend of Dirichlet and Neumann condition.

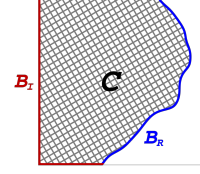


Figure 2: Cell cut into internal and Robin boundaries

4.6 Robin Velocity Update

The previous section determined the values of the A matrix, but the solution for p_+ must also be used when performing the velocity update. The result is a mixture of (16) and (17).

$$u'_+ = u_+ + \text{lerp}(j_+, q, \alpha) + (1 - \alpha)p_0 - p_0 \quad (28)$$

$$u'_+ = u_+ + \text{lerp}(j_+, q, \alpha) - \alpha p_0 \quad (29)$$

The result is a blend of the pressure and velocity rules.

5 Cut-cell Interpretation

Given the intermediate fluid velocity field $\mathbf{u}^*$, pressure projection finds a scalar field p such that $\nabla \cdot (\mathbf{u}^* - \nabla p) = 0$ everywhere in the domain, subject to boundary conditions. The two boundary conditions commonly used are Dirichlet ($p = q$) and Neumann ($(\mathbf{u}^* - \nabla p) \cdot \mathbf{n} = \mathbf{v} \cdot \mathbf{n}$, where $\mathbf{n}$ represents the boundary normal). Instead of treating them as disjoint, we can apply both simultaneously:

$$\alpha(p - q) + \beta(\mathbf{v} \cdot \mathbf{n} - (\mathbf{u}^* - \nabla p) \cdot \mathbf{n}) = 0, \quad (30)$$

where α, β are weighting factors capturing the relative importance of each condition.

Put into the form of the Robin condition:

$$\alpha p + \beta \frac{\partial p}{\partial \mathbf{n}} = \alpha q + \beta(\mathbf{u}^* - \mathbf{v}) \cdot \mathbf{n}. \quad (31)$$

Consider the cut-cell treatment of this problem [Papac et al. 2010]. Intersecting the fluid region with a cell near the boundary, we end up with a region C bounded by $\partial C = B_I \cup B_R$, where B_I are internal faces and B_R is where the Robin conditions must be satisfied (Figure 2). Then, by the divergence theorem, we have

$$0 = \iint_C \nabla \cdot (\mathbf{u}^* - \nabla p) dV = \int_{\partial C} (\mathbf{u}^* - \nabla p) \cdot d\mathbf{n} \quad (32)$$

$$0 = \int_{B_I} (\mathbf{u}^* - \nabla p) \cdot d\mathbf{n} + \int_{B_R} (\mathbf{u}^* - \nabla p) \cdot d\mathbf{n} \quad (33)$$

Assuming $\beta \neq 0$, equation 30 implies that

$$(\mathbf{u}^* - \nabla p) \cdot \mathbf{n} = \mathbf{v} \cdot \mathbf{n} + \frac{\alpha}{\beta}(p - q) \quad (34)$$

on B_R . Therefore,

$$0 = \int_{B_I} \mathbf{u}^* \cdot d\mathbf{n} - \int_{B_I} \nabla p \cdot d\mathbf{n} + \int_{B_R} \mathbf{v} \cdot d\mathbf{n} + \frac{\alpha}{\beta} \int_{B_R} (p - q) dA \quad (35)$$

Multiplying by β and isolating forms the pressure equation

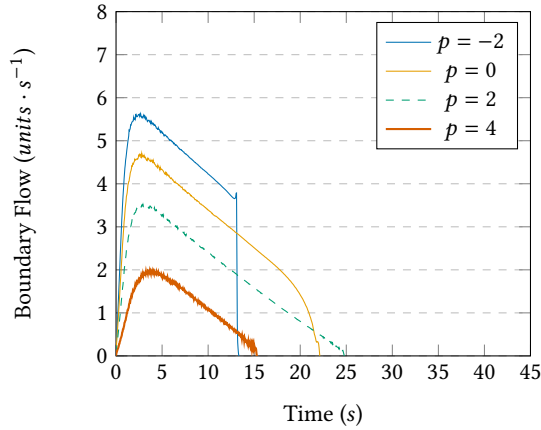


Figure 3: Pressure Boundary

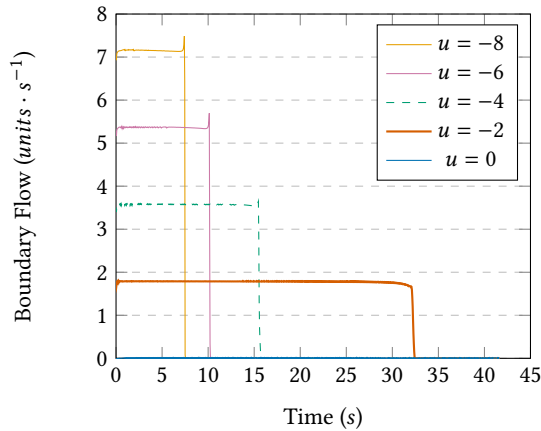


Figure 4: Velocity Boundary

$$-\beta \int_{B_I} \nabla p \cdot \mathbf{dn} + \alpha \int_{B_R} p dA = -\beta \int_{B_I} \mathbf{u}^* \cdot \mathbf{dn} - \beta \int_{B_R} \mathbf{v} \cdot \mathbf{dn} + \alpha \int_{B_R} q dA \quad (36)$$

or if $\mathbf{w}^*$ is the pre-enforced velocity field

$$-\beta \int_{B_I} \nabla p \cdot \mathbf{dn} + \alpha \int_{B_R} p dA = -\beta \iint_C \nabla \cdot \mathbf{w}^* dV + \alpha \int_{B_R} q dA \quad (37)$$

6 Results

We implemented these boundaries as a simple PCG-based liquid solver in a compositing network in Houdini 20.5. We modeled a 2d tank with a drain and graphed the vertical fluid flow above the sink. We stopped recording when flow stopped or reversed to avoid issues with bubbles collapsing.

The graphs validate the belief that the Robin boundary is qualitatively different. A sink with a velocity boundary achieves that velocity and maintains it until emptying. A sink with a pressure boundary will accelerate into the drain reaching a maximal velocity,

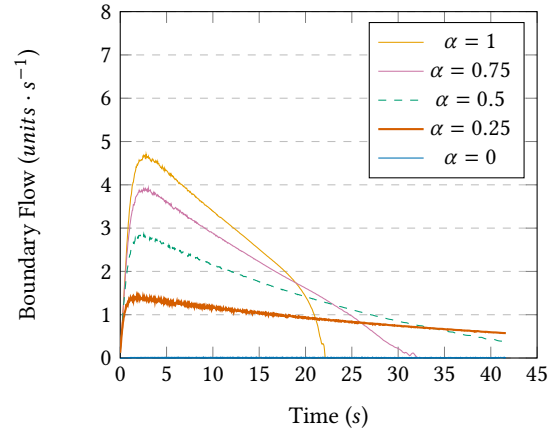


Figure 5: Robin Boundary: Zero Pressure, Zero Velocity

then slow as the head pressure decreases, until either the tank empties or the back-pressure is balanced. The Robin boundary allows a smooth blend between these options and produces unique velocity profiles. The slope for the pressure boundary is the same for all sink pressures, but mixing in the velocity condition with Robin boundaries produces different slopes.

The Robin condition is easy to add to existing simulators and can produce velocity profiles not achievable with traditional border conditions by controlling the porosity of free surfaces.

References

- Christopher Batty, Florence Bertails, and Robert Bridson. 2007. A fast variational framework for accurate solid-fluid coupling. *ACM Trans. Graph.* 26, 3 (2007), 100. <https://doi.org/10.1145/1276377.1276502>
- Ryan Goldade, Mridul Aanjaneya, and Christopher Batty. 2020. Constraint bubbles and affine regions: reduced fluid models for efficient immersed bubbles and flexible spatial coarsening. *ACM Trans. Graph.* 39, 4, Article 43 (aug 2020), 15 pages. <https://doi.org/10.1145/3386569.3392455>
- Jeff Lait. 2018. Divergence projection with electrostatics. In *ACM SIGGRAPH 2018 Talks* (Vancouver, British Columbia, Canada) (SIGGRAPH '18). Association for Computing Machinery, New York, NY, USA, Article 3, 2 pages. <https://doi.org/10.1145/3214745.3214752>
- Bailey Miller, Rohan Sawhney, Keenan Crane, and Ioannis Gkioulekas. 2024. Walkin' Robin: Walk on Stars with Robin Boundary Conditions. *ACM Trans. Graph.* 43, 4, Article 41 (jul 2024), 18 pages. <https://doi.org/10.1145/3658153>
- Joseph Papac, Frédéric Gibou, and Christian Ratsch. 2010. Efficient symmetric discretization for the Poisson, heat and Stefan-type problems with Robin boundary conditions. *J. Comput. Phys.* 229, 3 (2010), 875–889. <https://doi.org/10.1016/j.jcp.2009.10.017>
- Side Effects Software Inc. 2023. *Houdini 20.0*.
- Jos Stam. 1999. Stable Fluids. In *Proceedings of the 26th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '99)*. ACM Press/Addison-Wesley Publishing Co., New York, NY, USA, 121–128. <https://doi.org/10.1145/311535.311548>
- Alexey Stomakhin and Andrew Selle. 2017. Fluxed animated boundary method. *ACM Trans. Graph.* 36, 4, Article 68 (jul 2017), 8 pages. <https://doi.org/10.1145/3072959.3073597>